

DAYYER, Iran, Islamic Republic of

WMO: 408720

Lat: 27.833N

Lon: 51.933E

Elev: 13

StdP: 14.69

Time Zone: 3.50 (IRN)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 1             | 51.9       | 54.6 | 20.8                          | 15.6 | 65.3 | 25.9 | 19.9 | 66.0 | 27.9                  | 58.0 | 24.7 | 65.1 | 12.4                  | 310  | N/A |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.2                   | 111.4           | 72.7 | 107.9 | 73.1 | 104.7 | 73.6 | 89.6                | 97.3 | 88.7 | 96.7 | 87.9 | 96.2 | 13.9                 | 310  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 87.8                            | 202.3 | 96.3 | 86.8 | 195.5 | 95.7 | 85.8 | 189.5 | 95.2 | 55.7          | 97.5 | 54.3 | 96.7 | 53.3 | 96.2 | 93.2              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |                    | Extreme Annual Temperature |           |      |            | n-Year Return Period Values of Extreme Temperature |            |      |            |      |      |      |      |      |  |
|-------------------|------|--------------------|----------------------------|-----------|------|------------|----------------------------------------------------|------------|------|------------|------|------|------|------|------|--|
| Mean              |      | Standard Deviation |                            | n=5 years |      | n=10 years |                                                    | n=20 years |      | n=50 years |      |      |      |      |      |  |
| 1%                | 2.5% | 5%                 | Min                        | Max       | Min  | Max        | Min                                                | Max        | Min  | Max        | Min  | Max  |      |      |      |  |
| (a)               | (b)  | (c)                | (e)                        | (f)       | (g)  | (h)        | (i)                                                | (j)        | (k)  | (l)        | (m)  | (n)  | (o)  | (p)  |      |  |
| 23.1              | 20.2 | 17.8               | N/A                        | N/A       | N/A  | N/A        | N/A                                                | N/A        | N/A  | N/A        | N/A  | N/A  | N/A  | N/A  |      |  |
|                   |      |                    | DB                         |           |      |            |                                                    |            |      |            |      |      |      |      |      |  |
|                   |      |                    | WB                         | 39.2      | 90.9 | 3.6        | 1.3                                                | 36.6       | 91.8 | 34.5       | 92.6 | 32.5 | 93.4 | 29.9 | 94.3 |  |

Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 81.8   | 64.9  | 65.8  | 72.6  | 80.7  | 90.7  | 93.5  | 95.0  | 94.3  | 91.8  | 85.9  | 76.5  | 68.7  |
|                                                                        | DBStd       | 12.01  | 4.03  | 4.49  | 4.51  | 5.13  | 6.07  | 5.86  | 4.19  | 2.48  | 3.28  | 4.14  | 5.03  | 3.84  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 101    | 51    | 38    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 9     |
|                                                                        | CDD50       | 11605  | 462   | 443   | 701   | 920   | 1261  | 1306  | 1394  | 1375  | 1254  | 1113  | 796   | 580   |
|                                                                        | CDD65       | 6231   | 48    | 61    | 239   | 470   | 796   | 856   | 929   | 910   | 804   | 648   | 346   | 124   |
|                                                                        | CDH74       | 89760  | 72    | 135   | 1387  | 4912  | 12122 | 13875 | 15593 | 15488 | 13302 | 9260  | 3098  | 516   |
|                                                                        | CDH80       | 56996  | 1     | 9     | 278   | 2007  | 7741  | 9559  | 11129 | 11024 | 8988  | 5161  | 1050  | 49    |
| (14) Wind                                                              | WSAvg       | 7.4    | 8.8   | 8.5   | 7.9   | 7.2   | 8.3   | 8.7   | 6.7   | 5.9   | 6.5   | 6.2   | 7.3   | 7.5   |
| (15) Precipitation                                                     | PrecAvg     | 9.0    | 3.3   | 0.8   | 0.8   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 1.1   | 2.7   |
|                                                                        | PrecMax     | 19.7   | 10.9  | 2.5   | 3.2   | 0.6   | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.9   | 3.5   | 10.0  |
|                                                                        | PrecMin     | 3.7    | 0.4   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   |
|                                                                        | PrecStd     | 4.1    | 2.9   | 0.7   | 0.9   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 1.3   | 3.3   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 78.2  | 81.5  | 91.7  | 101.1 | 113.1 | 116.8 | 115.7 | 108.6 | 108.7 | 105.6 | 92.8  | 83.7  |
|                                                                        |             | MCWB   | 62.2  | 61.7  | 61.8  | 69.4  | 71.9  | 73.5  | 75.4  | 79.1  | 71.9  | 70.4  | 68.1  | 67.6  |
|                                                                        | 2%          | DB     | 75.9  | 77.4  | 86.0  | 96.4  | 108.8 | 111.7 | 110.6 | 104.4 | 103.3 | 100.5 | 89.5  | 81.1  |
|                                                                        |             | MCWB   | 64.9  | 63.6  | 64.2  | 69.0  | 70.4  | 71.7  | 74.8  | 82.3  | 73.8  | 72.2  | 69.4  | 66.2  |
|                                                                        | 5%          | DB     | 74.4  | 75.4  | 82.8  | 92.9  | 105.1 | 107.7 | 105.5 | 101.3 | 99.9  | 96.4  | 87.7  | 78.9  |
|                                                                        |             | MCWB   | 64.6  | 63.8  | 64.7  | 68.1  | 69.9  | 72.2  | 75.3  | 83.6  | 78.7  | 75.1  | 70.3  | 65.6  |
|                                                                        | 10%         | DB     | 73.0  | 73.7  | 80.4  | 89.4  | 101.8 | 103.9 | 101.4 | 99.4  | 98.1  | 94.0  | 85.8  | 77.0  |
|                                                                        |             | MCWB   | 63.2  | 62.8  | 64.3  | 68.3  | 70.2  | 72.7  | 78.5  | 85.4  | 82.2  | 76.8  | 70.0  | 63.3  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 69.8  | 69.2  | 73.3  | 78.5  | 83.8  | 88.2  | 90.0  | 91.5  | 89.5  | 86.0  | 78.1  | 72.9  |
|                                                                        |             | MCDB   | 75.2  | 75.1  | 81.0  | 85.5  | 89.8  | 94.5  | 96.9  | 99.0  | 97.5  | 93.6  | 86.0  | 80.6  |
|                                                                        | 2%          | WB     | 68.4  | 67.8  | 71.2  | 76.8  | 82.1  | 86.5  | 89.0  | 90.1  | 88.0  | 83.9  | 76.2  | 69.9  |
|                                                                        |             | MCDB   | 74.4  | 74.1  | 78.6  | 85.1  | 89.7  | 93.1  | 96.4  | 98.0  | 96.3  | 92.7  | 85.3  | 78.3  |
|                                                                        | 5%          | WB     | 66.9  | 66.5  | 70.0  | 74.9  | 80.8  | 85.5  | 88.2  | 89.1  | 87.0  | 82.4  | 74.0  | 67.7  |
|                                                                        |             | MCDB   | 73.2  | 73.1  | 77.3  | 84.1  | 89.5  | 92.3  | 95.9  | 97.3  | 95.8  | 91.8  | 84.3  | 76.7  |
|                                                                        | 10%         | WB     | 65.0  | 64.9  | 68.9  | 73.4  | 79.4  | 84.5  | 87.5  | 88.3  | 86.1  | 80.8  | 71.5  | 65.7  |
|                                                                        |             | MCDB   | 71.6  | 71.8  | 76.7  | 83.8  | 89.1  | 91.9  | 95.4  | 96.8  | 95.3  | 90.9  | 82.7  | 74.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.9  | 13.7  | 14.0  | 14.2  | 15.0  | 14.1  | 11.2  | 10.3  | 13.0  | 15.3  | 15.1  | 15.1  |
|                                                                        |             | MCDBR  | 14.3  | 14.3  | 17.0  | 18.7  | 20.9  | 20.9  | 18.7  | 14.0  | 15.6  | 18.4  | 16.1  | 16.4  |
|                                                                        |             | MCWBR  | 10.0  | 10.4  | 11.6  | 10.6  | 9.9   | 10.7  | 12.2  | 12.6  | 12.3  | 12.6  | 10.4  | 10.6  |
|                                                                        | 5% WB       | MCDBR  | 12.3  | 11.7  | 13.0  | 13.6  | 13.8  | 12.7  | 9.6   | 10.2  | 11.3  | 14.2  | 13.9  | 14.8  |
|                                                                        |             | MCWBR  | 11.1  | 10.6  | 12.5  | 12.0  | 13.6  | 12.3  | 10.2  | 9.9   | 10.9  | 12.3  | 11.9  | 11.3  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.467 | 0.527 | 0.591 | 0.656 | 0.641 | 0.621 | 0.707 | 0.674 | 0.580 | 0.525 | 0.480 | 0.454 |
|                                                                        | taud        |        | 1.987 | 1.842 | 1.694 | 1.609 | 1.627 | 1.681 | 1.574 | 1.647 | 1.780 | 1.881 | 1.988 | 2.030 |
|                                                                        | Ebn at Noon |        | 238   | 234   | 228   | 218   | 222   | 225   | 206   | 212   | 227   | 231   | 232   | 235   |
|                                                                        | Edh at Noon |        | 49    | 61    | 75    | 84    | 83    | 78    | 87    | 80    | 68    | 58    | 49    | 45    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1206  | 1458  | 1823  | 2086  | 2466  | 2627  | 2459  | 2311  | 2106  | 1752  | 1295  | 1135  |
| (45)                                                                   | RadStd      |        | 89    | 114   | 150   | 141   | 68    | 42    | 55    | 52    | 43    | 57    | 92    | 102   |

Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |